INCH-POUND
MIL-PRF-32383/5A
16 December 2021

SUPERSEDING
MIL-PRF-32383/5
29 February 2020

PERFORMANCE SPECIFICATION SHEET
BATTERY, RECHARGEABLE, SEALED, BB-2590B/U

The requirements for acquiring the product herein shall consist of this specification sheet and MIL-PRF-32383.

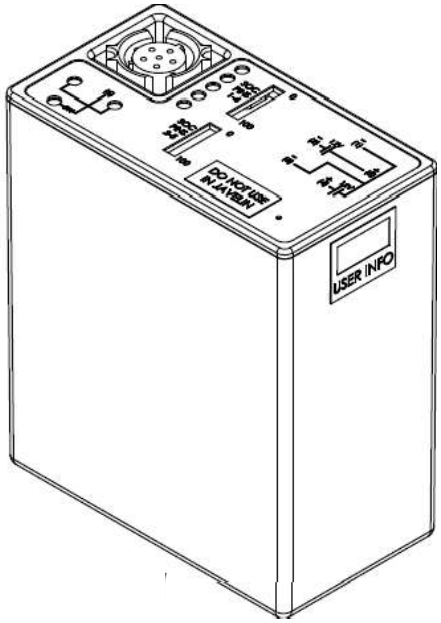


FIGURE 1. BB-2590B/U

Note: Depicted battery is representative of a potential solution but is not a requirement within this document. Unless otherwise specified, paragraph numbers are in reference to MIL-PRF-32383.

For clarity, MIL-PRF-32383 tables are labeled with capital Roman numerals (e.g. IX) and MIL-PRF-32383/5A tables are labeled with lower case Roman numerals (e.g. ix).

REQUIREMENTS.

Type – III (see 1.2 of MIL-PRF-32383 for classification) PIN – M32383/5- 3 (see 1.2 of MIL-PRF-32383 for PIN)

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Specification requirements and test methods: The following requirements and test methods of MIL-PRF-32383, identified therein as “when specified”, are applicable as indicated below:

TABLE i. *"When specified" requirements*

Requirement	Requirement Paragraph	Test Method Paragraph	Applicability
Additional material requirements	*3.3.7*	*4.7.1.9*	*No*
Connector pull force	*3.4.4.1.1.1*	*Certification*	*No*
Connector insertions	3.4.4.1.1.2	4.7.1.3	Yes
Multilayer finish	3.4.4.1.3	Certification	Yes
Gold plating	*3.4.4.1.4*	*Certification*	*No*
USB power port connector(s)	*3.4.4.1.5*	*4.7.2.17*	*No*
USB type A connector(s)	*3.4.4.1.5.1*	*4.7.2.17*	*No*
USB type C connector(s)	*3.4.4.1.5.2*	*4.7.2.17*	*No*
Surface friction	3.4.5.6	4.7.1.8	Yes
Item unique identification (IUID)	*3.4.6.1.1*	*4.7.1.1*	*No*
Additional physical characteristics requirements	*3.4.8*	*4.7.1.10*	*No*
Inrush current	3.5.5	4.7.2.12	Yes
Charge enable	3.5.6.2	4.7.2.6	Yes
Display, non-smart, always on	3.5.8.1.1	4.7.2.14.1	Yes
Display, non-smart, user activated	*3.5.8.1.2*	*4.7.2.14.1*	*No*
Display, smart	*3.5.8.2*	*4.7.2.14.2*	*No*
SMBus	3.5.9.1	4.7.2.15.1	Yes
USB power delivery	*3.5.9.2*	*4.7.2.15.2*	*No*
Dual-role power device	*3.5.9.2.1*	*4.7.2.14.2*	*No*
Concurrent charge/discharge	*3.5.11*	*4.7.2.16*	*No*
Additional electrical requirements	*3.5.12*	*4.7.2.18*	*No*
Vibration, loose cargo transportation	*3.6.5*	*4.7.3.7.1*	*No*
Temperature shock high altitude high opening (HAHO) simulation	*3.6.3*	*4.7.3.5.1*	*No*
Temperature shock high altitude low opening (HALO) simulation	*3.6.3*	*4.7.3.5.2*	*No*
Immersion, shallow	3.6.6	4.7.3.8.1	Yes
Immersion, deep	*3.6.6*	*4.7.3.8.2*	*No*
Transit drop, normal	*3.6.7*	*4.7.3.9.1*	*No*
Transit drop, severe	3.6.7	4.7.3.9.2	Yes
Drop while mated	*3.6.7*	*4.7.3.9.2*	*No*
Impact	*3.6.9*	*4.7.3.11*	*No*
Solar radiation	*3.6.10*	*4.7.3.12*	*No*
Salt fog	*3.6.11*	*4.7.3.13*	*No*
Electrostatic discharge (ESD)	*3.6.12*	*4.7.3.14*	*No*

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TABLE i. "When specified" requirements- Continued

Requirement	Requirement Paragraph	Test Method Paragraph	Applicability
Chemical resistance	*3.6.13*	*4.7.3.15*	*No*
Rapid decompression	*3.6.14*	*4.7.3.16*	*No*
Explosive atmosphere	*3.6.15*	*4.7.3.17*	*No*
Combined temperature, altitude, and humidity	*3.6.16*	*4.7.3.18*	*No*
Additional environmental requirements	*3.6.17*	*4.7.3.19*	*No*
Nail penetration, cell	3.7.1.2	4.7.4.4	Yes
Crush, cell	3.7.1.3	4.7.4.5	Yes
Inter-cell protective devices	*3.7.2.6*	*Certification*	*No*
Redundant safety protection	*3.7.2.7*	*Certification*	*No*
Projectile	3.7.2.8	4.7.4.11	Yes
Nail penetration, battery	*3.7.2.9*	*4.7.4.12*	*No*
Lithium battery safety program (US Navy)	3.7.2.10	4.7.5	Yes
Weapons characterization	*3.7.2.11*	*4.7.4.13*	*No*
Additional safety requirements	*3.7.3*	*4.7.4.14*	*No*
Electromagnetic interference	*3.8*	*4.7.7*	*No*
Additional transportation requirements	*3.9.1*	*4.7.8*	*No*
Reliability	*3.11*	*4.7.2.4.1*	*No*

1. Dimension and weight (3.4.3).
Dimensions: See FIGURE 2
Maximum weight: 3.25lb

2. Connector (3.4.4.1.1).
Connector: Drawing SC-C-179495 (or equivalent).
The face of connector shall be flush to .031 inch below top of battery Mating connector: N/A
Multilayer finish: Flat contacts shall be brass with multilayer finish Gold plating: N/A
Battery charger adapter: J-6B/P

3. Battery case vent (3.4.5.3).
See FIGURE 2 for location.

4. Markings (3.4.6).
See FIGURE 2 for the location of required markings as indicated. The locations for other required markings from MIL-PRF-32383 are optional.

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5. Instructions/notes (3.4.6.5).
See FIGURE 2.

6. Simplified battery label (3.4.6.6).
The simplified battery label location is optional. It shall be sized to remain legible.

7. Battery condition for shipping (3.4.7).
The instruction card shall at a minimum contain operation, storage, handling, maintenance, and warranty (if applicable) information.

8. Battery voltage (3.5.2).
Battery open circuit voltage:

Minimum		Maximum	
Single / Parallel	Series	Single / Parallel	Series
10 V	20 V	16.8 V	33.6 V

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9. Capacity (3.5.3).

Cutoff voltage: 20 V (series mode)

Each battery, discharged at the specified rate to minimal final voltage, shall have the following minimum capacity:

Test	Requirement Paragraph	Test Method Paragraph	Discharge Rate (A)	Minimum Capacity (Ah)
Capacity discharge	3.5.3	4.7.2.3	2	9.8
Cycle life (224th cycle)	3.5.4	4.7.2.4	2	9.3
High rate discharge	3.5.3	4.7.2.5	10	9.8
Low temperature discharge	3.5.3	4.7.2.7	2	8.9
High temperature discharge	3.5.3	4.7.2.8	2	9.8
Inrush current	3.5.5	4.7.2.12	See 3.5.5	9.7
Charge acceptance	3.5.10.1	4.7.2.9	2	5.1
Retention of charge	3.5.10.2	4.7.2.10	2	9.7
Pulse discharge[1]	3.5.3	4.7.2.11	See 4.7.2.11	19.5
Extreme low temperature discharge	3.6.1	4.7.3.2	2	8.6
Extreme high temperature discharge	3.6.1	4.7.3.3	2	9.8
Temperature shock (post- test)	3.6.3	4.7.3.5	2	9.8
Mechanical shock (post-test)	3.6.4	4.7.3.6	2	9.8
Vibration (during test)	3.6.5	4.7.3.7	0.8	9.7
Battery storage life	3.6.8	4.7.3.10	2	9.3
Overcharge/electric leakage (post-test)	3.7.2.2	4.7.4.7	2	9.8
Short circuit protection (post-test)	3.7.2.3	4.7.4.8	2	9.7
High temperature temporary cut off	3.7.2.4	4.7.4.9	2	9.7

NOTES

[1] Parallel mode; cutoff voltage is 10 V

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10. Inrush current (3.5.5).

The discharge profile shall be no less than as follows.

Step	Start time (minute:second:millisecond)	End time (minute:second:millisecond)	Current (A)
1	00:00:000	**05**:00:000	5
1a	**05**:00:000	05:00:**000.2**	60
2	05:00:**000.2**	05:00:**010**	22.5
3	05:00:**010**	05:00:**020**	20
4	05:00:**020**	05:00:**030**	19
5	05:00:**030**	05:00:**040**	18
6	05:00:**040**	05:00:**060**	17
7	05:00:**060**	05:00:**100**	16
8	05:00:**100**	05:00:**175**	15
9	05:00:**175**	05:00:**325**	14
10	05:00:**325**	05:00:**575**	13
11	05:00:**575**	05:00:**850**	12
12	05:00:**850**	05:**01**:000	11
13	05:**01**:000	**10**:00:000	10
14	Repeat steps 1 through 13		
15	**20**:00:010	Until cutoff voltage	5

11. Charging interface (3.5.6.1).

See FIGURE 2.

12. Charge enable (3.5.6.2).

The charge enable terminal shall comply with the following:

a. Maximum charge without enable: 400 mA
b. Equivalent resistor: 235 Ω
c. Equivalent diode V_F: 1.3 V
d. Approximate activation current: 7 mA

13. High temperature temporary cut off devices (3.7.2.4).

Not less than four devices

14. High temperature permanent cut off devices (3.7.2.5).

Not less than two devices

15. First article inspection (4.4).

Groups 1, 3, 4, and 5 apply

h. Group 5 (GR5) – Nail penetration tests (battery level and Weapons Characterization tests – samples size of six (6).

16. Constant potential, current limited (4.6.1).

16.8 V shall be applied to each battery section independently with current limited to 3 A, to a charge cut off of 100 mA.

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17. Alternate charging (4.6.3).
b. Charge each section independently with a constant potential of 16.8 V and current limit no greater than 3 A.

18. Capacity discharge (4.7.2.3).
c. 2 A.

19. Cycle life test (4.7.2.4).
a. and b. 2 A.

20. Pulse discharge (4.7.2.11).
d. Battery sections connected in parallel, 36 A for 5 seconds followed by 25 seconds off.

21. Nail penetration, cell (4.7.4.4).
SOC not less than 50%.

22. Overcharge/electric leakage (4.7.4.7).
c. Charge in parallel mode at 4 A with a charge voltage limit of 40 V.
f. 2 A.

ADDITIONAL FIGURES

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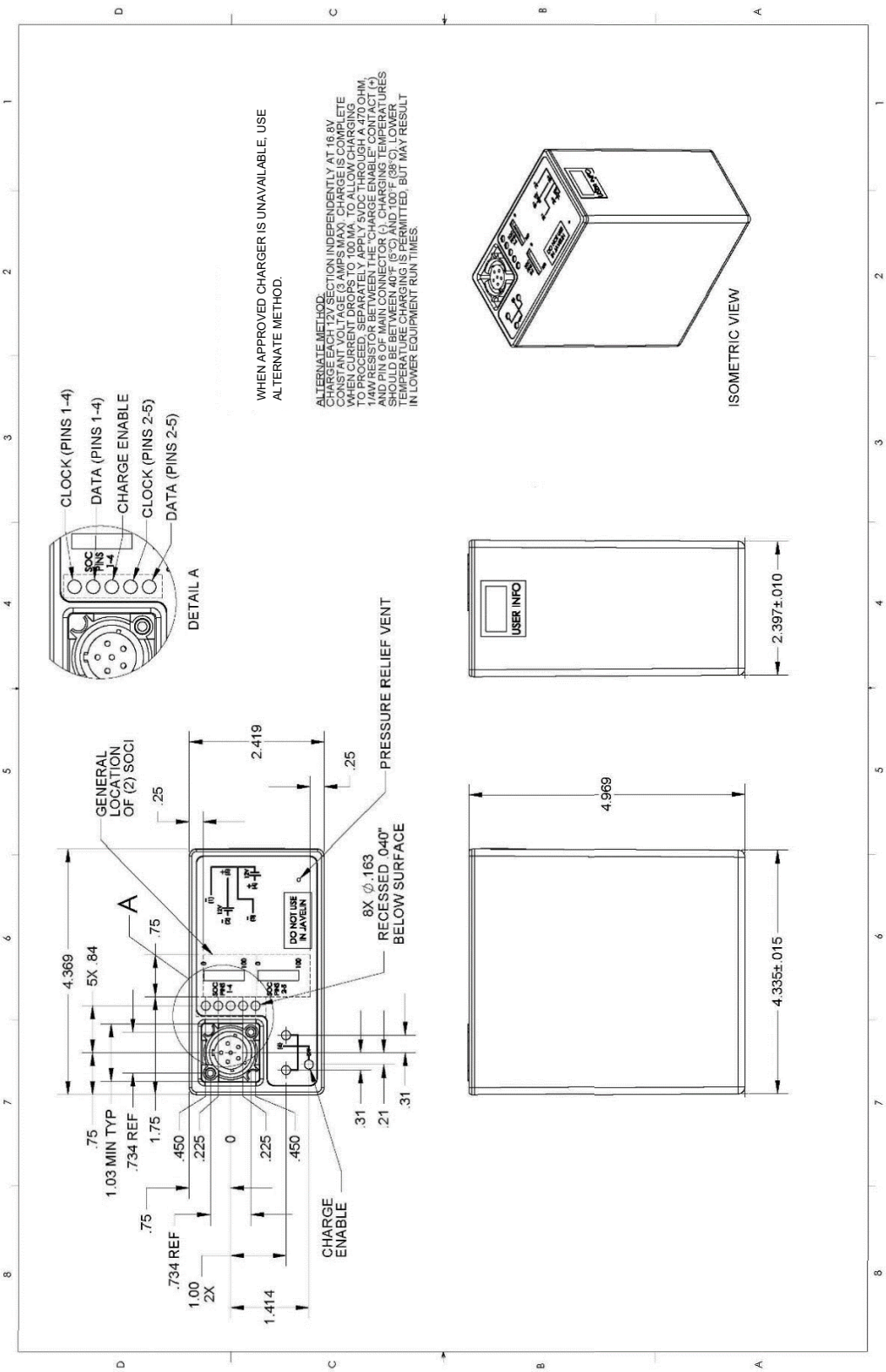


FIGURE 2. BB-2590B/U

NOTES

Dimensions are in inches. Tolerance is ± 0.031 inches unless stated otherwise.

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NOTES: (This section contains information of a general nature which may be helpful, but is not mandatory)

Nominal ratings: The following are the nominal ratings for the battery described by this specification sheet. They are provided for information purposes.

Battery PIN: M32383/5	**-3**
Type Classification:	BB-2590()/U
Chemistry:	Li-Ion
Color:	"Desert tan" Chip number: 33446
Weight (max):	3.25 lbs
Voltage Range:	20-33 (20-33.6[1]) V
Nominal Voltage:	28.8 V
Nominal Capacity:	9.6 (10.3[1]) Ah
Nominal Energy (new battery):	275 (295[1]) Wh
Rated power output:	≤ 280 W
Battery Life:	≥ 224 cycles, ≥ 3 years
Continuous load rating:	≤ 10 A
Pulse load rating:	≤ 36 A (5 SEC)
Charge temperature range:	-4 to 122 ºF (-20 to 50 ºC)
Operating temperature range:	-22 to 140 ºF (-30 to 55 ºC)
Storage temperature range:	-4 to 122 ºF (-20 to 50 ºC)
MAX abusive temperature: (non-operating):	199 ºF (93 ºC)
Connector:	SC-C-179495
Overall Dimensions:	~ 5 in. x 2.5 in. x 4.4 in.
Non-rechargeable equivalent / Specification:	BA-5590()/U, BA-5390()/U and BA-5790()/U / MIL-PRF-32271/1

NOTES

[1] When charged to 16.8 V

Other data:

Navy safety tests of NAVSEA S9310-AQ-SAF-010 are required for the BB-2590B/U batteries during first article testing. The test data provided from the testing will be used to evaluate applications for specific using devices and Navy platforms. Devices using this battery will require US Navy Safety Approval prior to use by Department of Navy users unless such approval has already been granted. Please consult with NAVSEA Instruction 9310.1 for further information.

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Expected performance with Legacy Chargers (16.5 V):

Test	Requirement Paragraph	Test Method Paragraph	Discharge Rate (A)	Minimum Capacity (Ah)
Capacity discharge	3.5.3	4.7.2.3	2	9.5
Cycle life (224th cycle)	3.5.4	4.7.2.4	2	8.5
High rate discharge	3.5.3	4.7.2.5	10	9.3
Low temperature discharge	3.5.3	4.7.2.7	2	8.4
High temperature discharge	3.5.3	4.7.2.8	2	9.5
Inrush current	3.5.5	4.7.2.12	See 3.5.5	9
Charge acceptance	3.5.10.1	4.7.2.9	2	4.8
Retention of charge	3.5.10.2	4.7.2.10	2	9
Pulse discharge 1/	3.5.3	4.7.2.11	See 4.7.2.11	18.5
Extreme low temperature discharge	3.6.1	4.7.3.2	2	8.4
Extreme high temperature discharge	3.6.1	4.7.3.3	2	9.5
Temperature shock (post- test)	3.6.3	4.7.3.5	2	9.5
Mechanical shock (post-test)	3.6.4	4.7.3.6	2	9.5
Vibration (during test)	3.6.5	4.7.3.7	0.8	9.5
Battery storage life	3.6.8	4.7.3.10	2	8.9
Overcharge/electric leakage (post-test)	3.7.2.2	4.7.4.7	2	9.5
Short circuit protection (post-test)	3.7.2.3	4.7.4.8	2	9
High temperature temporary cut off	3.7.2.4	4.7.4.9	2	9

Changes from previous issue: Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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Army – CR
Navy – MC
Air Force – 71
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Preparing activity:
Army – CR
(Project Number 6140-2021-016)

Review activities:
Navy – AS, SH

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